

(Originally-submitted appendix results)

H The Shared MLP Addition Module

H.1 Identifying Addition Neurons

Figure 42 shows the distribution of write scores for all layer 18 MLP neurons (Section 5.1): this score measures the proportion of a neuron’s down projection row $\mathbf{d}_i$ that is within the best DAS output subspace at layer 18. We choose a threshold $\tau = 0.4$ by eye based on the addition task, and find that neurons in all other tasks are a subset of these 28 addition neurons. Table 6 gives change in performance when this set of 28 neurons is ablated, as well as when all other neurons at layer 18 are ablated. These addition neurons can explain most of the important computation at the layer 18 MLP.

Neuron ablations. We show ablation results for our canonical tasks in Table 6. We also test ablation on unseen templates in Tables 7, 8, 9, and 10, finding that these neurons are also important for rephrasings of the same task. Figure 43 shows errors for Llama-3.1-8B on the addition task, compared to errors when all MLP neurons at layer 18 are zero-ablated except for $\mathcal{N}_{\text{add}}$; failure for larger numbers suggests that we may be missing larger period neurons.

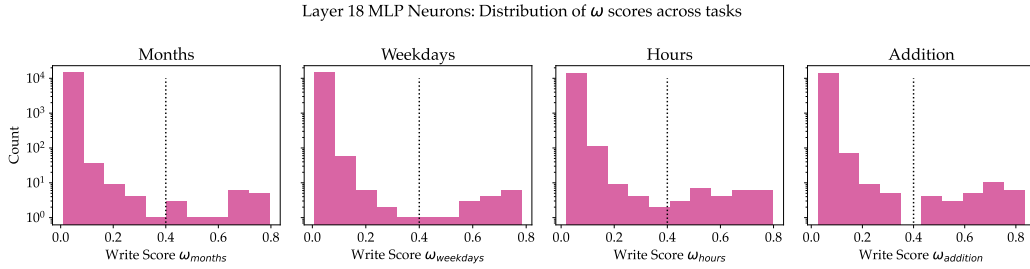


Figure 42: Distribution of write scores ω for layer 18 MLP neurons. A higher score means that this neuron’s down projection row $\mathbf{d}_i$ is within the best output DAS subspace for this task. We pick the threshold $\tau = 0.4$ by eye based on the addition task. There are 28 neurons with $\omega > 0.4$ for the addition task, 16 for months, 15 for weekdays, and 26 for hours. All neurons for cyclic tasks are contained within the set of addition neurons, except for a single hours neuron.

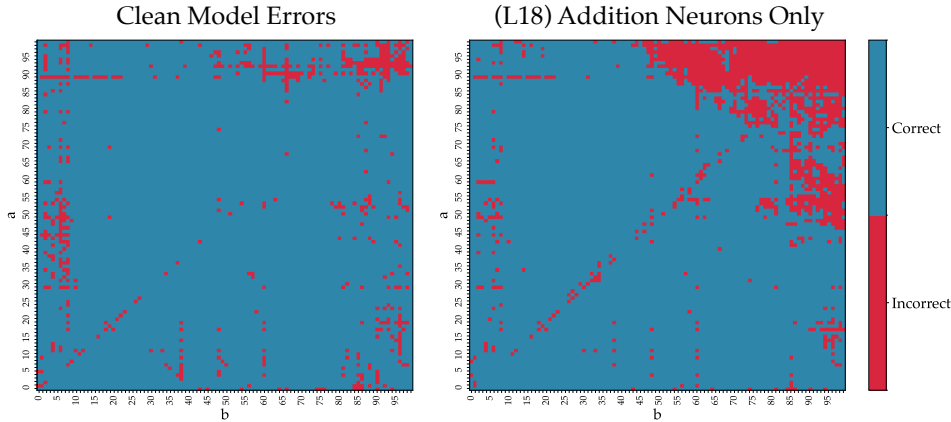


Figure 43: Llama-3.1-8B errors on the addition task. We show errors for a clean model run (95% accuracy), as well as errors when all neurons at the L18 MLP are zero-ablated except for our 28 addition neurons $\mathcal{N}_{\text{add}}$ (86% accuracy). Most errors come from higher number ranges, suggesting that $\mathcal{N}_{\text{add}}$ excludes some neurons with larger periods.

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Table 6: Accuracy (%) intervening on the set of 28 addition neurons $\mathcal{N}_{\text{add}}$. **Clean**: Llama-3.1-8B accuracy on this task. **Only** $\mathcal{N}_{\text{add}}$: accuracy when all other neuron activations at the layer 18 MLP are set to 0, except for neurons in $\mathcal{N}_{\text{add}}$. Despite these neurons making up 0.2% of layer 18 neurons, accuracy remains high. **Zeroed**: accuracy drops significantly when these neurons’ activations are set to zero. **Flipped**: reversing the sign of these neurons’ activations is destructive, likely because they represent periodic functions that oscillate between negative and positive values across inputs.

Task	Number Range	Clean	Only $\mathcal{N}_{\text{add}}$	Zeroed	Flipped	Chance
Weekdays	1→7	91.84	89.80	40.82	20.41	14.29
	8→14	68.37	58.16	28.57	17.35	14.29
Months	1→12	81.25	77.78	36.81	14.58	8.33
	13→24	64.93	70.83	23.61	16.32	8.33
Hours	1→24	93.06	80.56	50.69	16.32	4.17
	24→48	69.18	51.65	29.77	11.63	4.17
Addition	$a, b \in [1..100]$	94.60	86.36	24.11	2.97	0.50

Rebuttal Result 1: Extra Ablations for 16 New Templates (4 per task)

Table 7: Neuron intervention results for the addition task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	Input Format	Offset Format	Clean	Only ↑	Zeroed ↓	Flipped ↓	Random ↑
(default prompt)	1, 2...	1, 2...	94.33%	86.11%	23.80%	2.92%	94.47% ± 0.17
The sum of {input} and {offset} equals	1, 2...	1, 2...	100.00%	98.87%	42.74%	5.09%	100.0% ± 0.00
Math question: {input} plus {offset} equals	1, 2...	1, 2...	96.76%	76.98%	29.30%	4.22%	96.66% ± 0.32
Q: What is the sum of {input} and {offset}? A:	1, 2...	1, 2...	99.70%	97.79%	43.93%	5.41%	99.68% ± 0.06
Q: If I have {input} apples and buy {offset} more, how many apples do I have? Total apples:	1, 2...	1, 2...	75.67%	40.63%	12.32%	1.43%	74.92% ± 2.47

Rebuttal Result 1 cont.

Table 8: Neuron intervention results for the weekdays task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	Input Format	Offset Format	Clean	Only $\uparrow$	Zeroed $\downarrow$	Flipped $\downarrow$	Random $\uparrow$
(default prompt)	Monday, Tuesday...	one, two...	91.84%	89.80%	40.82%	20.41%	92.45% $\pm$ 1.31
Let's do some weekday math. {offset} days after {input} is what day? Answer:	Monday, Tuesday...	one, two...	73.47%	73.47%	30.61%	16.33%	73.47% $\pm$ 1.83
Question: Today is {input}. In {offset} days, what day will it be? Answer: It will be	Monday, Tuesday...	1, 2...	97.96%	85.71%	44.90%	24.49%	95.31% $\pm$ 1.84
If today is {input}, and I have a deadline in {offset} days, what day is my deadline? My deadline is on	Mon, Tues...	one, two...	83.67%	71.43%	40.82%	20.41%	86.53% $\pm$ 1.63
Let's do some calendar math. Today is {input}. What day will it be in {offset} days? Answer:	Mon, Tues...	1, 2...	85.71%	83.67%	38.78%	30.61%	83.47% $\pm$ 1.70

Rebuttal Result 1 cont.

Table 9: Neuron intervention results for the months task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	Input Format	Offset Format	Clean	Only $\uparrow$	Zeroed $\downarrow$	Flipped $\downarrow$	Random $\uparrow$
(default prompt)	January, February...	one, two...	81.25%	76.39%	36.81%	14.58%	81.32% $\pm$ 1.00
Let’s do some calendar math. If the current month is {input}, then in {offset} months, what month will it be? Answer: It will be	January, February...	one, two...	97.22%	96.53%	59.03%	22.22%	96.74% $\pm$ 0.82
Question: I have a big presentation in {offset} months. If it is currently {input}, then what month is my presentation? Answer: My presentation is in	January, February...	1, 2...	70.83%	64.58%	46.53%	25.00%	70.97% $\pm$ 0.52
Q: A pregnant woman will give birth in {offset} months. If it is currently {input}, what month will she give birth? Answer: She will give birth in	Jan, Feb...	one, two...	91.67%	89.58%	49.31%	17.36%	91.88% $\pm$ 1.28
Let’s figure out some scheduling. I just got my hair cut, and it is currently {input}. If I have to book another appointment in {offset} months, what month will it be? Answer: My appointment is in	Jan, Feb...	1, 2...	81.25%	78.47%	53.47%	34.03%	80.90% $\pm$ 1.33

Rebuttal Result 1 cont.

Table 10: Neuron intervention results for the hours task across alternative prompt templates, for one cycle each. **Input / Offset Format** columns show examples of strings input to model. **Clean**: unmodified accuracy for Llama-3.1-8B. **Only**: accuracy when all neurons at layer 18 except for $\mathcal{N}_{\text{add}}$ are zero-ablated. **Zeroed**: accuracy when all 28 $\mathcal{N}_{\text{add}}$ addition neurons are zero-ablated. **Flipped**: accuracy when 28 $\mathcal{N}_{\text{add}}$ neurons have their signs flipped. **Random**: accuracy when 28 random non-addition neurons at layer 18 are zero-ablated (averaged across 10 seeds).

Template	Input Format	Offset Format	Clean	Only $\uparrow$	Zeroed $\downarrow$	Flipped $\downarrow$	Random $\uparrow$
(default prompt)	00, 01...	one, two...	97.57%	84.55%	50.87%	16.15%	97.24% $\pm$ 0.28
Let's do some clock math. If it is {input}:00 right now in 24-hour time, then what time will it be in {offset} hours? Answer: When I was in the military, I learned 24-hour time. So if right now it is {input}:00, then in {offset} hours, it will be	00, 01...	one, two...	90.45%	68.40%	47.40%	13.37%	90.31% $\pm$ 1.30
Q: If it is {input} o' clock in military time, what time will it be in {offset} hours? A: In 24-hour time, it will be	00, 01...	one, two...	91.32%	83.68%	50.52%	18.06%	91.15% $\pm$ 0.28
I need to work out when to get to the airport. Q: It's {input}:00 right now in 24-hour time. My flight is in {offset} hours. What time is my flight? Answer: My flight will be at	00, 01...	one, two...	81.08%	73.26%	47.57%	23.26%	81.23% $\pm$ 0.31
	00, 01...	1, 2...	86.46%	64.58%	34.20%	12.67%	86.22% $\pm$ 0.59

Rebuttal Result 2: Sweep over neuron thresholds.

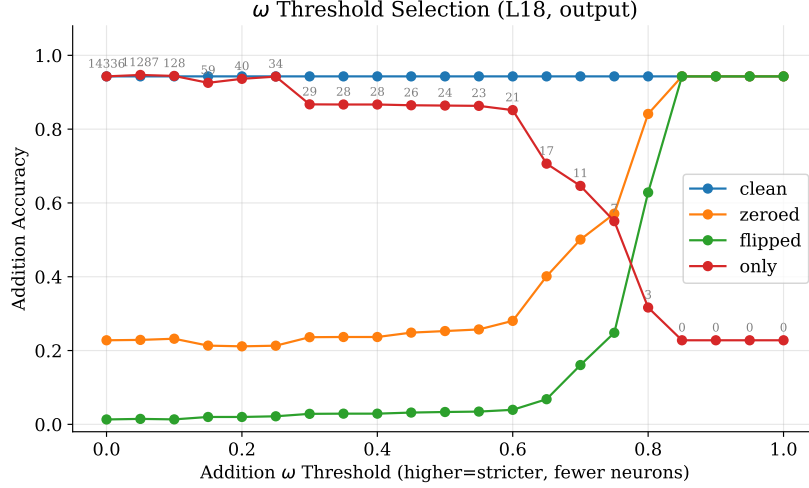


Figure 58: Sweeping over ω for the addition task. While we choose $\omega > 0.4$, keeping more neurons ($\omega > 0.2$) allows for cleaner results with still only 40 addition neurons. Gray text indicates the number of neurons included for this threshold. We include analysis on the additional neurons not in $\mathcal{N}_{add}$ in App. H.6.

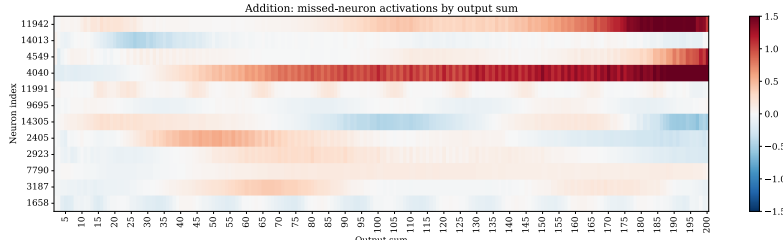


Figure 59: Activations for twelve extra neurons that are included when $\omega > 0.2$ for the addition task. These neurons also show wavelike patterns when activations are averaged across output sums.

H.6 Extra Addition Neurons

After sweeping over possible thresholds for ω for the addition task in Figure 58, we notice that cleaner ablation results can be achieved with a more lax threshold, including layer 18 MLP neurons with $\omega > 0.2$.

We plot the activation patterns for the additional twelve neurons (40 - 28) not yet analyzed in this paper in Figure 59, observing that they also have periodic patterns, with activations oscillating along output sums. In fact, neurons 11991 and 1658 appear to be missing period 20 neurons, while others appear to encode larger periods, as hypothesized in Figure 43 (where we observed that errors with our original set of 28 neurons were mostly for larger numbers).

Rebuttal Result 2: Sweep over neuron thresholds.

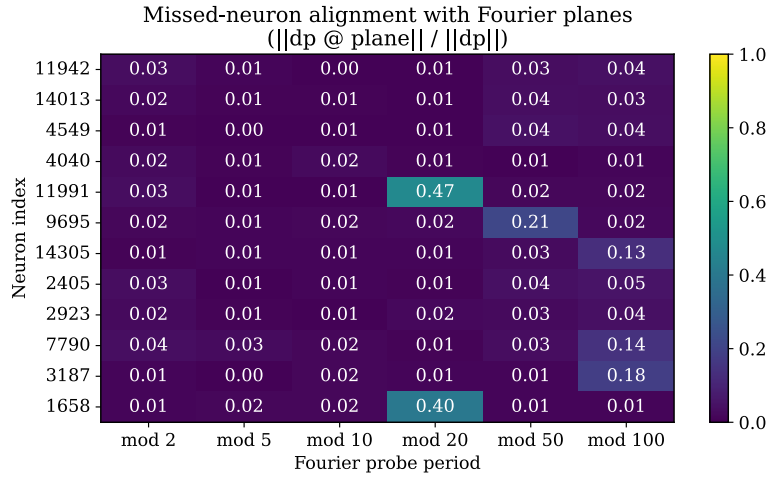


Figure 60: Alignment of twelve extra neurons from Figure 59 that are included when $\omega > 0.2$ for the addition task. Two neurons (11991, 1658) appear to be period 20 neurons that could be included in $\mathcal{N}_{add}$. Neuron 9695 appears aligned with $T = 50$, and neurons 14305, 7790, and 3187 are slightly aligned with $T = 100$. For other neurons, we see low alignment—we hypothesize that this is because these neurons write to larger periods. We can directly observe this in Figure 59: e.g., neuron 2405 is positive at 50 and negative at 200, suggesting that it has a period of $T = 300$, and neurons 11942 and 4040 appear to gradually activate for a large range of sums. Also see Figure 43.